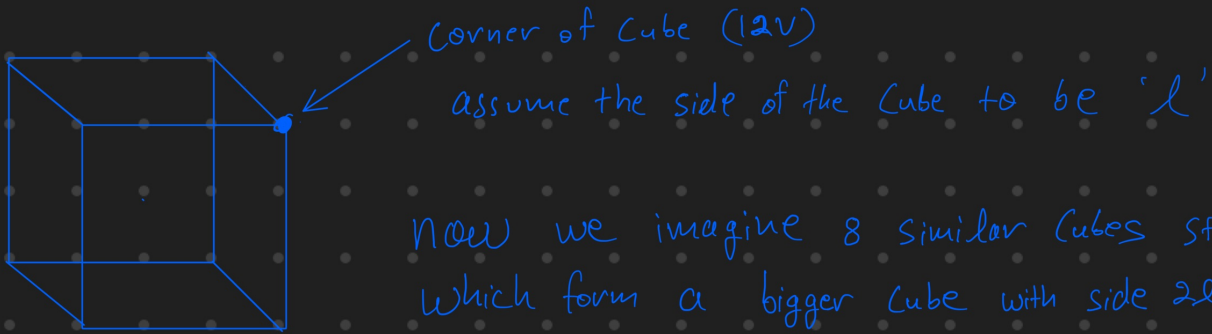
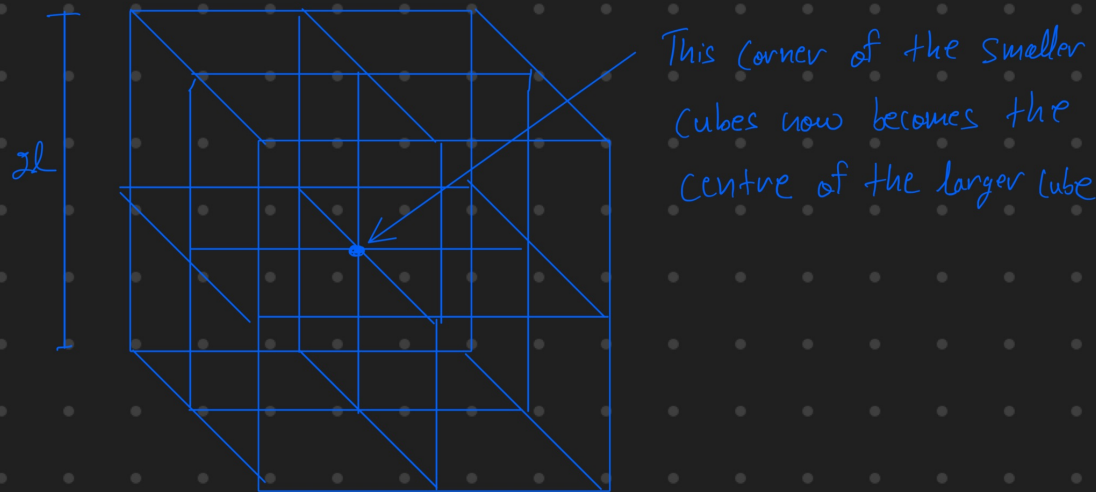


Q. The potential at the vertex of a cube uniformly charged throughout its volume is 12V. What is the potential at the centre of the cube?



Now we imagine 8 similar Cubes stacked together which form a bigger cube with side $2l$



Now if we compare the Cube with Side length 'l' to the one with side length ' $2l$ '. The potential at the centre of Cube ' $2l$ ' will be 4 times the potential at the Centre of the Cube with Side ' l '. As $V \propto l^2$

The potential at centre of bigger Cube will be $(12)8V = 96V$.
Now half of 96 volt is what will be the potential at the centre of the smaller Cube with side length ' l ' $\therefore V_{\text{centre}} = \frac{96}{4} = \boxed{24V}$

This method does not involve the use of Integration to find the potential.
This method is a benchmark for certain types of olympiad problems, Simple yet highly effective.